Fleet condition report: Penmanshiel wind farm

Penmanshiel wind farm (14x Senvion MM82, UK), Cubico, Greenbyte export. 14 turbines, SCADA 02 Jun 2016 to 31 Dec 2024; assessment window: the last 24 months of each turbine (Jan 2022 to Dec 2024). Generated by `windml report` from the mapping `penmanshiel_all_years.yaml`.

Executive summary

10 of 14 turbines are green, 3 amber, 1 red. Red means act now: a strong, specific warning sign. Amber means check at the next visit: a specific, unusual pattern. Green means no validated warning sign in the data, not a guarantee.

Turbine	Rating	Why	Suggested check	Confidence
PEN14	RED	Gear-oil pump pressure at full load is 2.3 bar above its own 12-month baseline (now 10.0 bar), while the other turbines are at +0.0 bar; flagged 3 months running.	Inspect the gearbox oil filter and pump: read the filter differential pressure, check the bypass and pressure-relief valve, take an oil sample for particles. The same build-up was seen before in PEN01 (+3.0 bar, Aug 2023); PEN02 (+2.7 bar, Jun 2023 followed by 3 oil-pressure trip episodes); PEN04 (+3.2 bar, Jul 2023 followed by 8 oil-pressure trip episodes).	High
PEN02	AMBER	Tripped on oil-system faults in 5 separate episodes in the last 6 months (other turbines: median 0); most often 'Missing gear oil (high rpm)' (5 trips); last on 23 Nov 2023. (This turbine's published data ends 31 Dec 2023.)	No trip in the last 30 days and pump pressure dropped 2.6 bar, which looks like a repair or oil/filter change: confirm it in the work orders. If not repaired: inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch.	High
PEN04	AMBER	Tripped on oil-system faults in 9 separate episodes in the last 6 months (other turbines: median 0); most often 'Low gearbox oil pressure' (10 trips); last on 28 Nov 2023. (This turbine's published data ends 31 Dec 2023.)	No trip in the last 30 days and pump pressure dropped 2.6 bar, which looks like a repair or oil/filter change: confirm it in the work orders. If not repaired: inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch.	High
PEN09	AMBER	Tripped on oil-system faults in 4 separate episodes in the last 6 months (other turbines: median 0); most often 'Missing gear oil (high rpm)' (22 trips); last on 05 Nov 2023. (This turbine's published data ends 31 Dec 2023.)	No trip in the last 30 days: confirm in the work orders that the cause was repaired. Inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch.	High

Green: PEN01, PEN05, PEN06, PEN07, PEN08, PEN10, PEN11, PEN12, PEN13, PEN15.

Worth knowing. Shown for context only: these signals (model alerts, past availability) did not reliably precede events in the validation, so they do not change the rating.

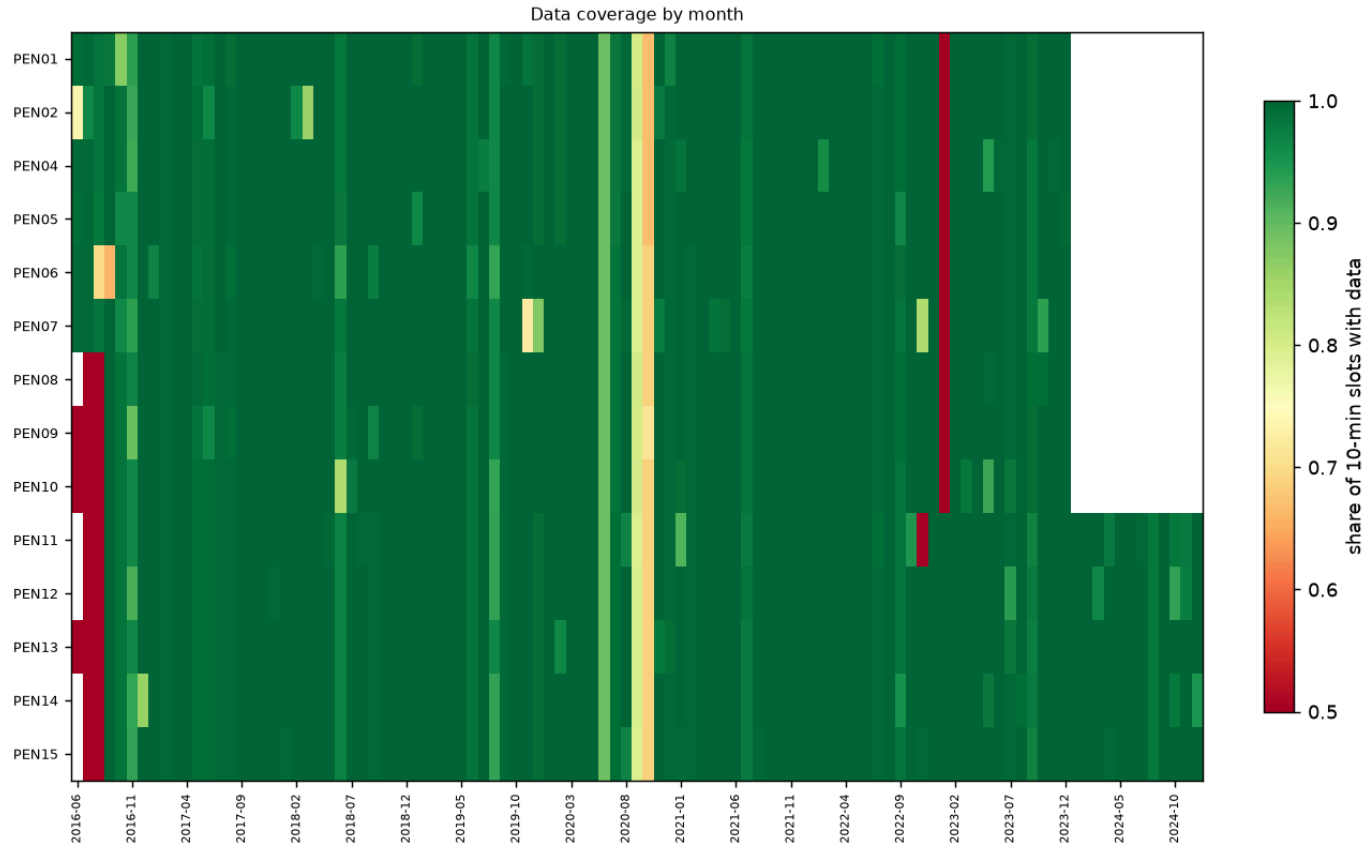
- **PEN01**: Its gearbox oil system behaved unlike its own normal-behaviour model on 62% of days in the last 3 months.
- **PEN02**: Its gearbox oil system behaved unlike its own normal-behaviour model on 62% of days in the last 3 months, probably because pump pressure dropped 2.6 bar (oil or filter change?).
- **PEN04**: Its gearbox oil system behaved unlike its own normal-behaviour model on 64% of days in the last 3 months, probably because pump pressure dropped 2.6 bar (oil or filter change?).
- **PEN07**: Availability was 94.4% over the last 6 months vs 99.0% for the other turbines; largest fault cause: hydraulics (164 h). This is past downtime, not a forecast.
- **PEN07**: Its pitch system behaved unlike its own normal-behaviour model on 45% of days in the last 3 months.
- **PEN14**: Its gearbox oil system behaved unlike its own normal-behaviour model on 34% of days in the last 3 months, in line with the pump-pressure build-up.
- **PEN15**: Its main bearing behaved unlike its own normal-behaviour model on 56% of days in the last 3 months.

How the ratings work. A turbine is amber only for a specific, unusual pattern: gear-oil pump pressure building up far beyond its sister turbines, repeated oil-system or pitch trips, or a faulty sensor. It is red when the pattern is strong (e.g. a pump pressure rise of 2 bar or more) or persistent and still active. Thresholds were chosen on 2017-2020 data from two UK wind farms and checked on 2021-2024: amber or red turbine-months were followed by a logged fault, trip or long service stop within 90 days 76% of the time, against 50% for random turbines in the same months; but only 19% of the turbine-months before such an event carried a flag. Ratings help decide where to look first; most faults still arrive without a warning sign in SCADA data.

How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years, an alarm log and at least 3 sister turbines to compare with; it drops when any of these is missing. A red rating is a reason to look, not a diagnosis.

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
PEN01	2016-06-06	2023-12-31	91	95.6%	744	9	10min	0	-	alarm log
PEN02	2016-06-03	2023-12-31	91	95.6%	744	9	10min	0	-	alarm log
PEN04	2016-06-13	2023-12-31	91	95.1%	744	9	10min	661	-	alarm log
PEN05	2016-06-15	2023-12-31	90	95.4%	744	9	10min	0	-	alarm log
PEN06	2016-06-02	2023-12-31	91	95.5%	744	9	10min	0	-	alarm log
PEN07	2016-06-02	2023-12-31	91	94.6%	744	9	10min	0	-	alarm log
PEN08	2016-07-27	2023-12-31	89	95.5%	744	9	10min	0	-	alarm log
PEN09	2016-06-24	2023-12-31	90	95.6%	744	9	10min	0	-	alarm log
PEN10	2016-06-27	2023-12-31	90	95.1%	744	9	10min	0	-	alarm log
PEN11	2016-07-19	2024-12-31	101	99.4%	16	9	10min	0	-	alarm log
PEN12	2016-07-02	2024-12-31	102	99.0%	50	9	10min	2	-	alarm log
PEN13	2016-06-30	2024-12-31	102	99.7%	10	9	10min	0	-	alarm log
PEN14	2016-07-09	2024-12-31	102	99.4%	22	9	10min	0	-	alarm log
PEN15	2016-07-14	2024-12-31	102	99.7%	9	9	10min	1,233	-	alarm log



Signals present in at least one turbine: `ambient_temp_c`, `available_capacity_kw`, `converter_inlet_temp_c`, `data_availability`, `energy_export_kwh`, `gear_oil_inlet_pressure_bar`, `gear_oil_inlet_temp_c`, `gear_oil_pump_pressure_bar`, `gear_oil_temp_c`, `gen_bearing_front_temp_c`, `gen_bearing_rear_temp_c`, `generator_speed_rpm`, `hub_temp_c`, `lost_prod_curtail_total_kwh`, `lost_prod_downtime_kwh`, `main_bearing_front_temp_c`, `main_bearing_rear_temp_c`, `nacelle_position_deg`, `nacelle_temp_c`, `pitch_a_deg`, `pitch_b_deg`, `pitch_c_deg`, `pitch_motor_temp_1_c`, `pitch_motor_temp_2_c`, `pitch_motor_temp_3_c`, `power_kw`, `power_setpoint_kw`, `rotor_bearing_temp_c`, `rotor_speed_rpm`, `wind_direction_deg`, `wind_speed_1_ms`, `wind_speed_2_ms`, `wind_speed_ms`, `yaw_bearing_angle_deg`.

2. Turbine scorecard

Turbine	Flag	Confidence	Tech. avail.	Energy avail.	Fault h/yr	Largest fault cause	Oil ep.	Pitch ep.	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
PEN01	GREEN	High	99.1%	97.6%	39	pitch (76 h)	2	3	38%	0%	7.2 to 7.5	1.1x
PEN02	AMBER	High	98.5%	97.5%	92	gearbox (108 h)	13	0	57%	0%	7.7 to 8.1	1.1x
PEN04	AMBER	High	94.2%	94.6%	262	unspecified (375 h)	11	1	56%	0%	7.7 to 8.0	0.9x
PEN05	GREEN	High	96.6%	94.6%	166	pitch (251 h)	1	1	12%	0%	7.8 to 8.0	0.9x
PEN06	GREEN	High	94.5%	91.9%	143	pitch (257 h)	4	5	3%	0%	7.5 to 7.7	1.0x
PEN07	GREEN	High	97.4%	95.4%	183	hydraulics (164 h)	5	4	14%	0%	7.0 to 7.8	1.2x
PEN08	GREEN	High	98.0%	96.5%	114	converter (154 h)	6	2	0%	0%	6.8 to 6.9	0.9x
PEN09	AMBER	High	98.6%	97.3%	87	gearbox (119 h)	7	0	1%	0%	6.8 to 6.9	0.9x
PEN10	GREEN	High	91.1%	92.4%	739	unspecified (1471 h)	0	3	0%	0%	6.7 to 7.0	0.9x
PEN11	GREEN	High	98.5%	95.7%	66	unspecified (37 h)	1	1	42%	0%	6.8 to 6.6	1.1x
PEN12	GREEN	High	98.0%	95.4%	116	pitch (77 h)	3	3	1%	0%	7.1 to 7.6	1.1x
PEN13	GREEN	High	99.5%	97.5%	6	tower (11 h)	0	0	1%	0%	7.3 to 7.6	1.2x
PEN14	RED	High	98.9%	97.3%	38	electrical (70 h)	0	0	8%	0%	7.8 to 9.9	1.4x
PEN15	GREEN	High	97.0%	95.4%	89	pitch (71 h)	16	5	22%	1%	7.4 to 6.9	3.1x

Ratings, reasons and notes

- **PEN01** GREEN (confidence High: 91 months of history, models trained on earlier years). No rating trigger. Notes: Its gearbox oil system behaved unlike its own normal-behaviour model on 62% of days in the last 3 months. 4.4% of its SCADA data is missing in the window.
- **PEN02** AMBER (confidence High: 91 months of history, models trained on earlier years). AMBER: Tripped on oil-system faults in 5 separate episodes in the last 6 months (other turbines: median 0); most often 'Missing gear oil (high rpm)' (5 trips); last on 23 Nov 2023. *Check:* No trip in the last 30 days and pump pressure dropped 2.6 bar, which looks like a repair or oil/filter change: confirm it in the work orders. If not repaired: inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch. Notes: Its gearbox oil system behaved unlike its own normal-behaviour model on 62% of days in the last 3 months, probably because pump pressure dropped 2.6 bar (oil or filter change?). 4.4% of its SCADA data is missing in the window.
- **PEN04** AMBER (confidence High: 91 months of history, models trained on earlier years). AMBER: Tripped on oil-system faults in 9 separate episodes in the last 6 months (other turbines: median 0); most often 'Low gearbox oil pressure' (10 trips); last on 28 Nov 2023. *Check:* No trip in the last 30 days and pump pressure dropped 2.6 bar, which looks like a repair or oil/filter change: confirm it in the work orders. If not repaired: inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch. Notes: Its gearbox oil system behaved unlike its own normal-behaviour model on 64% of days in the last 3 months, probably because pump pressure dropped 2.6 bar (oil or filter change?). 4.9% of its SCADA data is missing in the window.
- **PEN05** GREEN (confidence High: 90 months of history, models trained on earlier years). No rating trigger. Notes: 4.6% of its SCADA data is missing in the window.
- **PEN06** GREEN (confidence High: 91 months of history, models trained on earlier years). No rating trigger. Notes: 4.5% of its SCADA data is missing in the window.

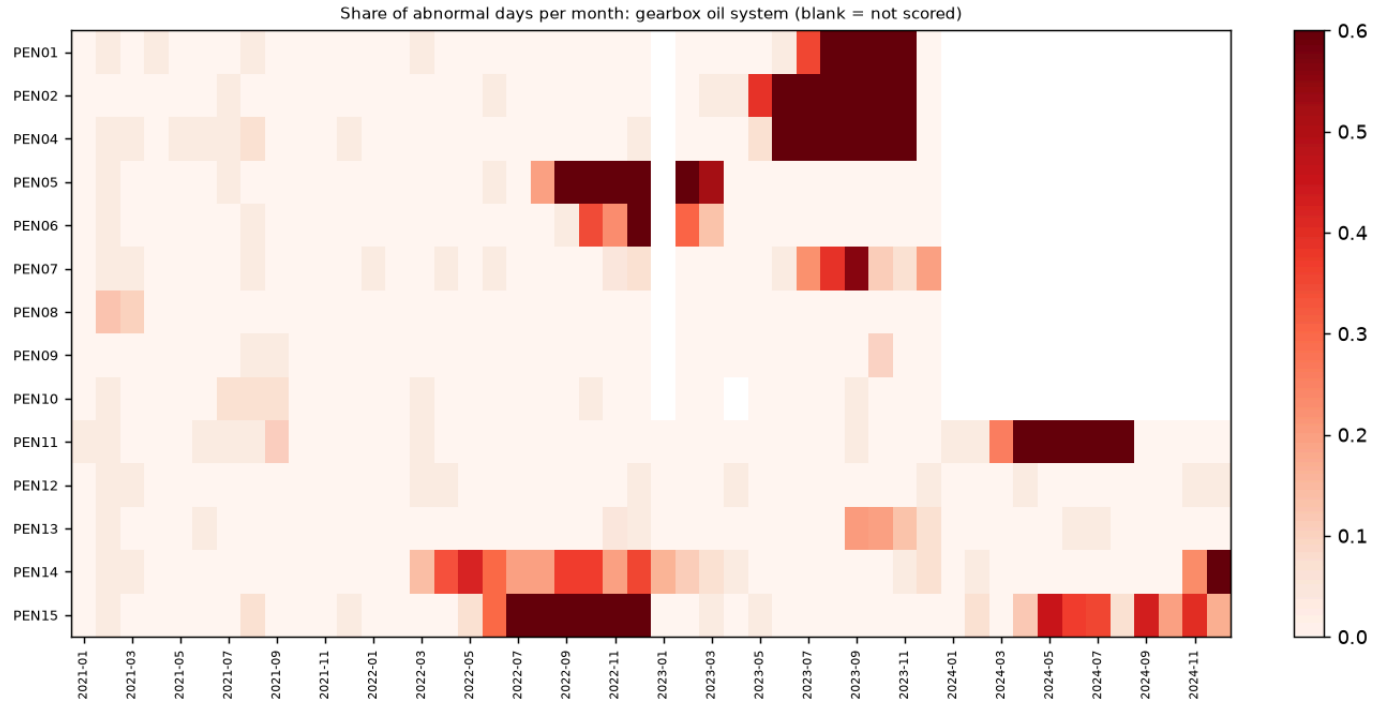
- **PEN07** GREEN (confidence High: 91 months of history, only 95% data coverage, models trained on earlier years). No rating trigger. Notes: Availability was 94.4% over the last 6 months vs 99.0% for the other turbines; largest fault cause: hydraulics (164 h). This is past downtime, not a forecast. Its pitch system behaved unlike its own normal-behaviour model on 45% of days in the last 3 months. 5.4% of its SCADA data is missing in the window.
- **PEN08** GREEN (confidence High: 89 months of history, models trained on earlier years). No rating trigger. Notes: 4.5% of its SCADA data is missing in the window.
- **PEN09** AMBER (confidence High: 90 months of history, models trained on earlier years). AMBER: Tripped on oil-system faults in 4 separate episodes in the last 6 months (other turbines: median 0); most often 'Missing gear oil (high rpm)' (22 trips); last on 05 Nov 2023. *Check:* No trip in the last 30 days: confirm in the work orders that the cause was repaired. Inspect the gearbox oil circuit: filter, pump, cooler, oil level and the pressure switch. Notes: 4.4% of its SCADA data is missing in the window.
- **PEN10** GREEN (confidence High: 90 months of history, models trained on earlier years). No rating trigger. Notes: 4.9% of its SCADA data is missing in the window.
- **PEN11** GREEN (confidence High: 101 months of history, models trained on earlier years). No rating trigger.
- **PEN12** GREEN (confidence High: 102 months of history, models trained on earlier years). No rating trigger.
- **PEN13** GREEN (confidence High: 102 months of history, models trained on earlier years). No rating trigger.
- **PEN14** RED (confidence High: 102 months of history, models trained on earlier years). RED: Gear-oil pump pressure at full load is 2.3 bar above its own 12-month baseline (now 10.0 bar), while the other turbines are at +0.0 bar; flagged 3 months running. *Check:* Inspect the gearbox oil filter and pump: read the filter differential pressure, check the bypass and pressure-relief valve, take an oil sample for particles. The same build-up was seen before in PEN01 (+3.0 bar, Aug 2023); PEN02 (+2.7 bar, Jun 2023 followed by 3 oil-pressure trip episodes); PEN04 (+3.2 bar, Jul 2023 followed by 8 oil-pressure trip episodes). Notes: Its gearbox oil system behaved unlike its own normal-behaviour model on 34% of days in the last 3 months, in line with the pump-pressure build-up.
- **PEN15** GREEN (confidence High: 102 months of history, models trained on earlier years). No rating trigger. Notes: Its main bearing behaved unlike its own normal-behaviour model on 56% of days in the last 3 months.

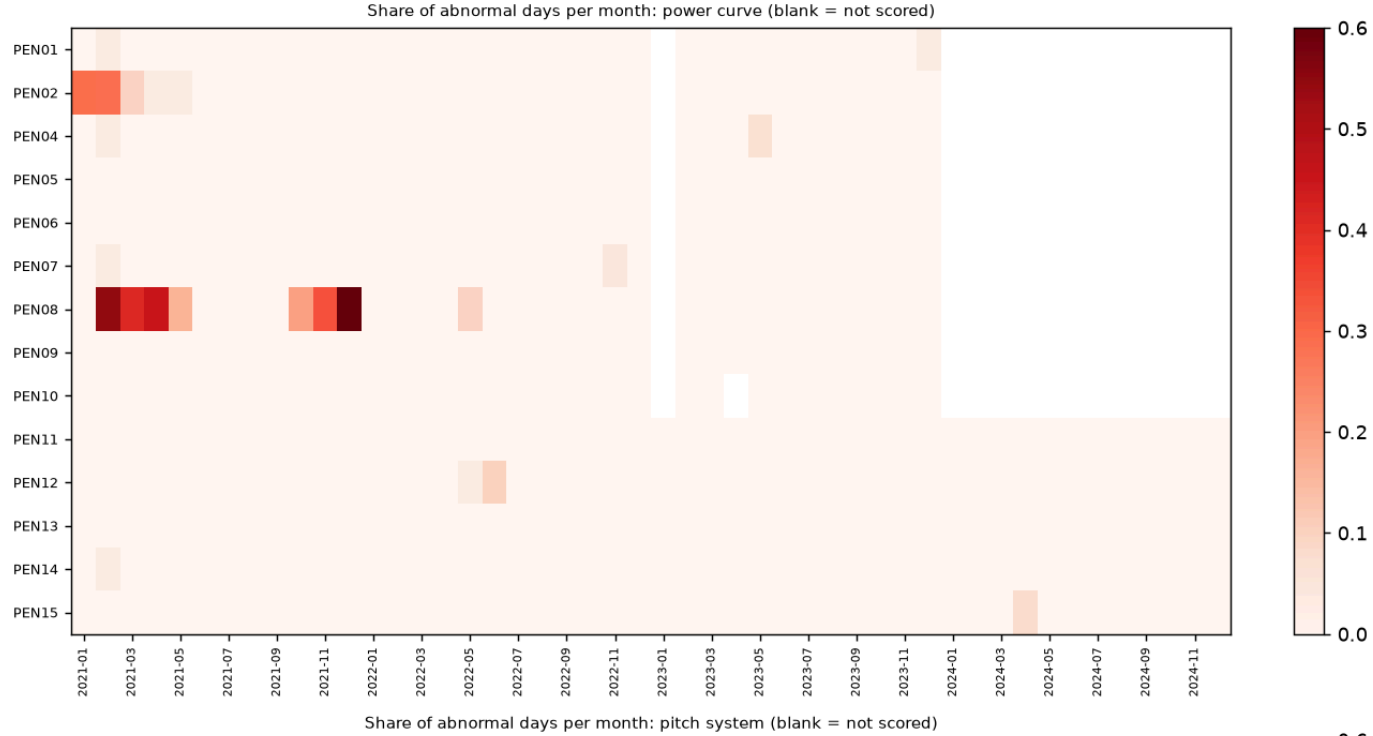
3. Model-based health (normal-behaviour residuals)

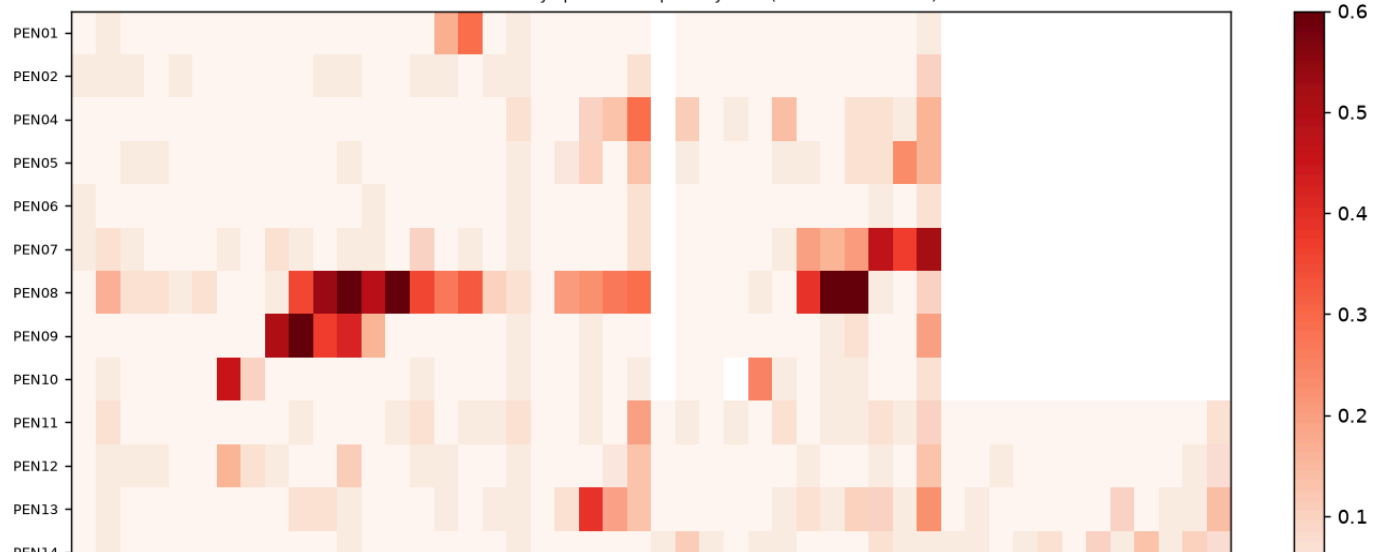
Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

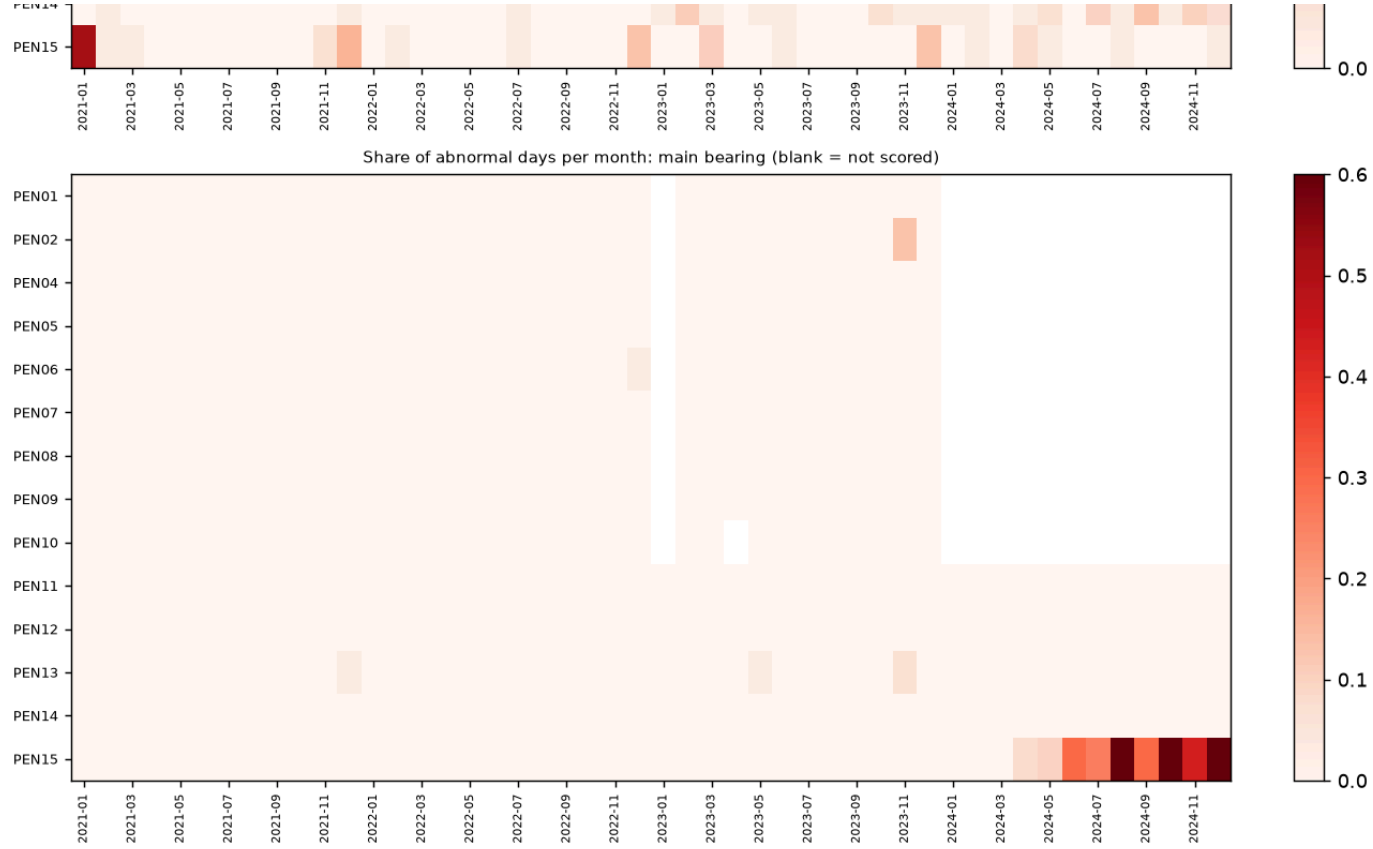
Turbine	Training scheme	Models	Window scored	gearbox oil system: abnormal days final yr / chronic months	main bearing: abnormal days final yr / chronic months	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
PEN01	walk-forward	14	95%	38% / 5	0% / 0	0% / 0	0% / 0
PEN02	walk-forward	14	94%	57% / 7	1% / 0	1% / 0	0% / 0
PEN04	walk-forward	14	90%	56% / 6	0% / 0	6% / 0	0% / 0
PEN05	walk-forward	14	92%	12% / 6	0% / 0	6% / 0	0% / 0
PEN06	walk-forward	14	91%	3% / 3	0% / 0	1% / 0	0% / 0
PEN07	walk-forward	14	94%	14% / 2	0% / 0	18% / 3	0% / 0
PEN08	walk-forward	14	95%	0% / 0	0% / 0	18% / 7	0% / 0
PEN09	walk-forward	14	95%	1% / 0	0% / 0	3% / 0	0% / 0
PEN10	walk-forward	14	87%	0% / 0	0% / 0	3% / 0	0% / 0
PEN11	walk-forward	14	99%	42% / 5	0% / 0	1% / 0	0% / 0
PEN12	walk-forward	14	99%	1% / 0	0% / 0	1% / 0	0% / 0
PEN13	walk-forward	14	100%	1% / 0	0% / 0	3% / 0	0% / 0
PEN14	walk-forward	14	99%	8% / 1	0% / 0	5% / 0	0% / 0
PEN15	walk-forward	14	98%	22% / 5	29% / 6	2% / 0	1% / 0

Chronic abnormal turbines (3+ consecutive months with >= 30% abnormal days): PEN01 gearbox oil system (5 months); PEN02 gearbox oil system (7 months); PEN04 gearbox oil system (6 months); PEN05 gearbox oil system (6 months); PEN07 pitch system (3 months); PEN08 pitch system (3 months); PEN11 gearbox oil system (5 months); PEN15 gearbox oil system (3 months); PEN15 main bearing (5 months).


Share of abnormal days per month: gearbox oil system (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
PEN14
PEN15
2021-01
2021-03
2021-05
2021-07
2021-09
2021-11
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0

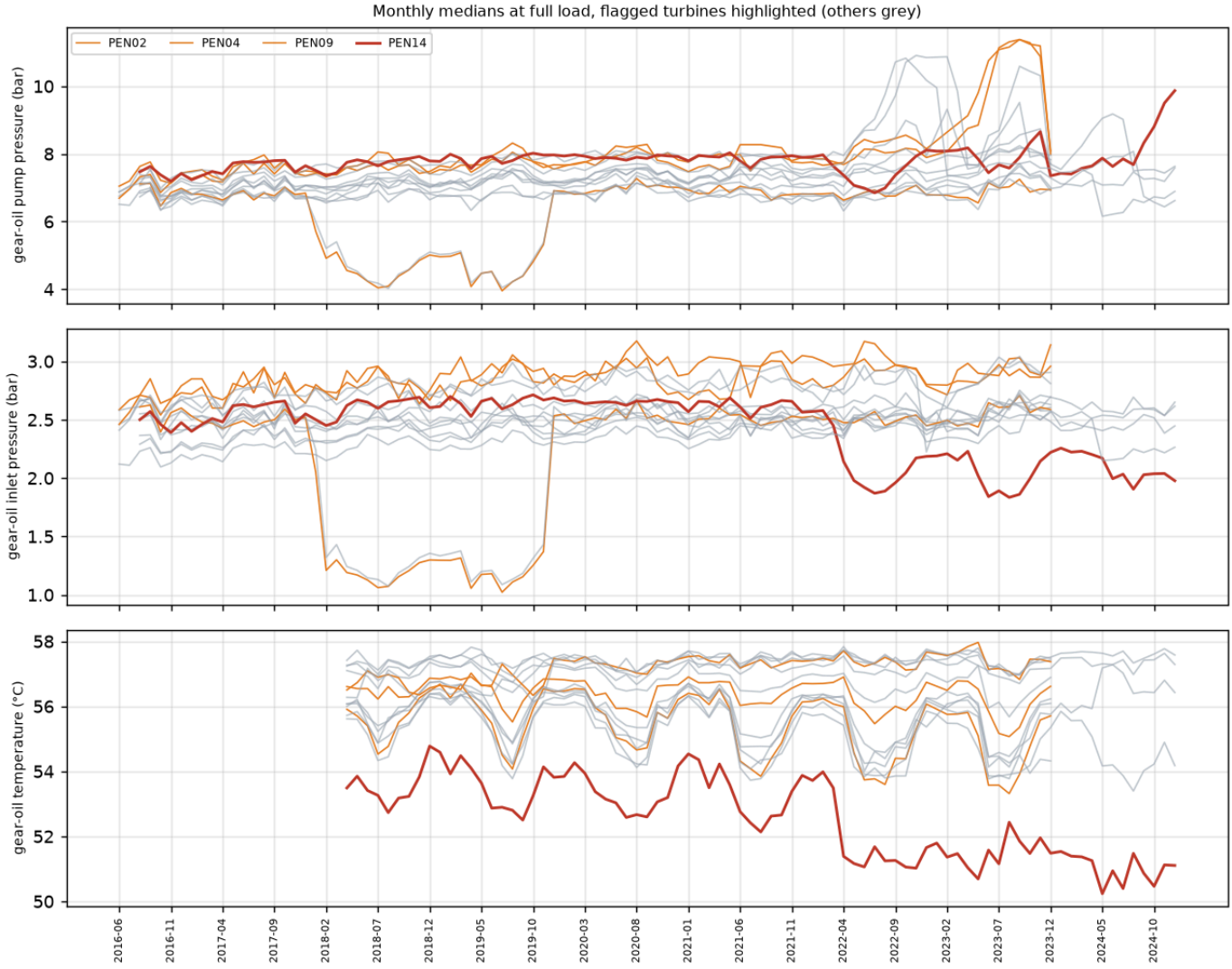

Share of abnormal days per month: power curve (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
PEN14
PEN15
2021-01
2021-03
2021-05
2021-07
2021-09
2021-11
2022-01
2022-03
2022-05
2022-07
2022-09
2022-11
2023-01
2023-03
2023-05
2023-07
2023-09
2023-11
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
0.6
0.5
0.4
0.3
0.2
0.1
0.0


Share of abnormal days per month: pitch system (blank = not scored)
PEN01
PEN02
PEN04
PEN05
PEN06
PEN07
PEN08
PEN09
PEN10
PEN11
PEN12
PEN13
0.6
0.5
0.4
0.3
0.2
0.1



4. Gearbox oil pressure and temperature trends

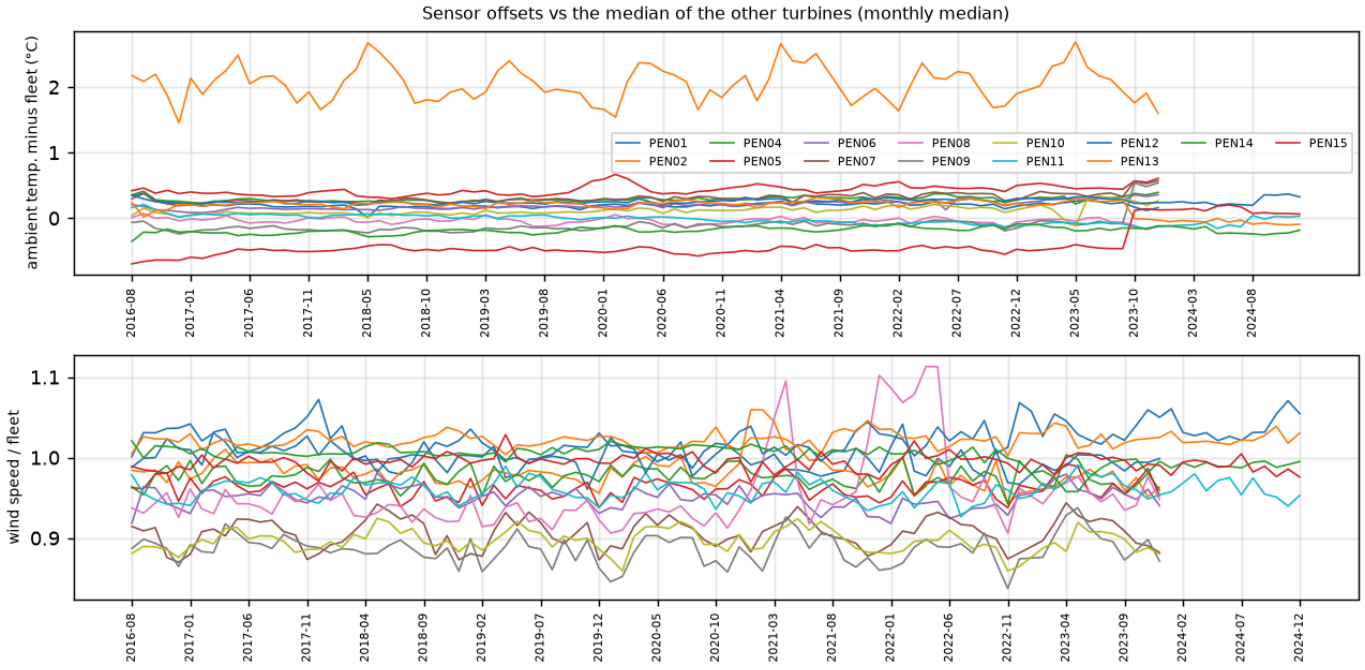
Turbine	Pump pressure (bar): baseline → last month	Inlet pressure (bar)	Oil temperature (°C)	Baseline months
PEN01	7.2 → 7.5 (+0.3)	2.5 → 2.7 (+0.2)	57.0 → 56.9 (-0.1)	2016-06..2021-12
PEN02	7.7 → 8.1 (+0.4)	2.8 → 3.0 (+0.1)	55.8 → 55.7 (-0.1)	2016-06..2021-12
PEN04	7.7 → 8.0 (+0.3)	2.9 → 3.1 (+0.3)	56.6 → 56.6 (+0.0)	2016-06..2021-12
PEN05	7.8 → 8.0 (+0.2)	2.8 → 2.9 (+0.1)	56.1 → 56.2 (+0.1)	2016-06..2021-12
PEN06	7.5 → 7.7 (+0.2)	2.8 → 2.8 (+0.0)	56.1 → 56.1 (+0.0)	2016-06..2021-12
PEN07	7.0 → 7.8 (+0.8)	2.3 → 2.6 (+0.3)	56.0 → 54.3 (-1.7)	2016-06..2021-12
PEN08	6.8 → 6.9 (+0.1)	2.5 → 2.6 (+0.1)	57.4 → 57.4 (+0.0)	2016-08..2021-12
PEN09	6.8 → 6.9 (+0.1)	2.5 → 2.6 (+0.1)	57.2 → 57.4 (+0.2)	2016-08..2021-12
PEN10	6.7 → 7.0 (+0.3)	2.5 → 2.6 (+0.1)	57.2 → 57.5 (+0.3)	2016-09..2021-12
PEN11	6.8 → 6.6 (-0.2)	2.5 → 2.4 (-0.0)	57.4 → 57.6 (+0.1)	2016-08..2022-12
PEN12	7.1 → 7.6 (+0.5)	2.6 → 2.6 (+0.0)	57.2 → 57.3 (+0.1)	2016-08..2022-12
PEN13	7.3 → 7.6 (+0.4)	2.4 → 2.6 (+0.2)	55.4 → 54.2 (-1.3)	2016-08..2022-12
PEN14	7.8 → 9.9 (+2.1)	2.6 → 2.0 (-0.6)	53.2 → 51.1 (-2.1)	2016-08..2022-12
PEN15	7.4 → 6.9 (-0.5)	2.4 → 2.3 (-0.1)	55.6 → 56.4 (+0.8)	2016-08..2022-12


Monthly medians at full load, flagged turbines highlighted (others grey)
PEN02
PEN04
PEN09
PEN14
gear-oil pump pressure (bar)
gear-oil inlet pressure (bar)
gear-oil temperature (°C)
2016-06
2016-11
2017-04
2017-09
2018-02
2018-07
2018-12
2019-05
2019-10
2020-03
2020-08
2021-01
2021-06
2021-11
2022-04
2022-09
2023-02
2023-07
2023-12
2024-05
2024-10

5. Sensor drift and stuck sensors

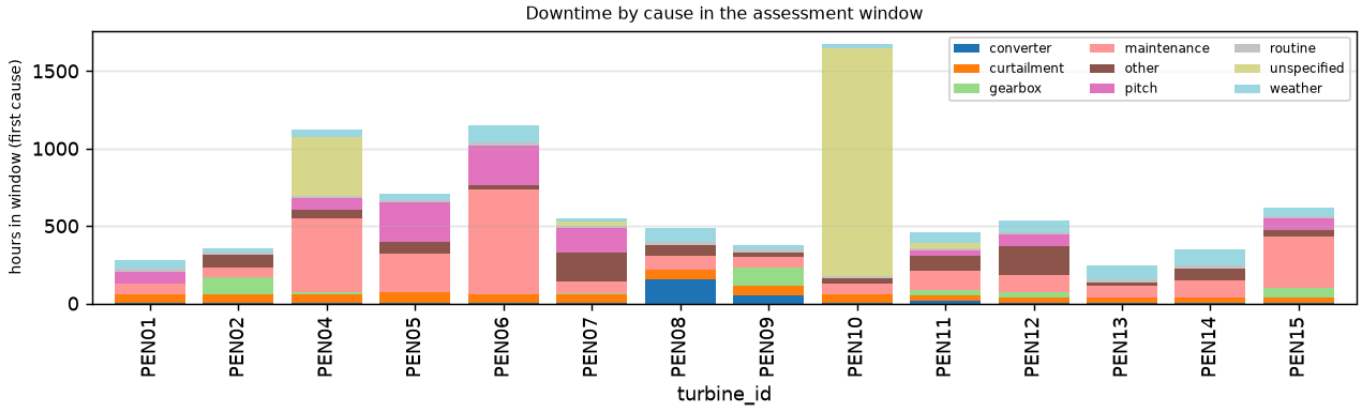
Drift = change of a turbine's offset to the median of the other turbines, last 3 months vs before the window (thresholds: 1.5 °C, 5% wind speed).

Turbine	drift ambient_temp_c	drift wind_speed_ms	Stuck sensors
PEN01	-0.03 °C	-0.6%	-
PEN02	-0.31 °C	-2.1%	-
PEN04	+0.10 °C	-1.1%	-
PEN05	+0.17 °C	0.7%	-
PEN06	+0.19 °C	-0.4%	-
PEN07	+0.28 °C	-1.8%	-
PEN08	-0.09 °C	1.1%	-
PEN09	+0.68 °C	-0.2%	-
PEN10	+0.17 °C	-1.5%	-
PEN11	+0.02 °C	-0.7%	-
PEN12	+0.10 °C	4.0%	-
PEN13	-0.32 °C	0.9%	-
PEN14	-0.04 °C	-1.6%	-
PEN15	+0.56 °C	-1.9%	-



6. Stop and fault history

Turbine	Fault stops	Maintenance	Grid	Weather	Curtailment	Major-fault episodes	Maint. h/yr	Oil / pitch trips
PEN01	24	45	1	102	15	2	34	2 / 12
PEN02	90	44	1	93	15	2	33	16 / 0
PEN04	89	44	0	80	15	4	236	12 / 2
PEN05	78	45	0	104	15	1	122	1 / 2
PEN06	54	49	1	111	16	3	336	4 / 9
PEN07	47	35	1	84	15	4	40	8 / 17
PEN08	205	46	2	115	14	7	44	9 / 3
PEN09	51	47	3	92	16	3	32	25 / 0
PEN10	25	34	27	94	16	1	35	0 / 4
PEN11	45	43	36	167	13	3	59	10 / 2
PEN12	99	46	4	173	14	2	56	6 / 3
PEN13	21	36	15	161	14	0	37	0 / 0
PEN14	33	38	3	196	14	0	56	0 / 0
PEN15	105	52	4	132	13	6	165	34 / 6



Most frequent fault stops in the window (all turbines):

Code	Message	Component	Stops
6530	Anemometer defect	sensor_met	269
4530	Tower oscillation Y level 2	tower	154
1550	Missing gear oil (high rpm)	gearbox	95
4510	Tower oscillation Y level 1	tower	91
3000	Frequency converter not ready	converter	86
3110	Frequency converter error	converter	45
3555	Current asymmetry	unspecified	31
4540	Tower oscillation X level 2	tower	19
1510	Low gearbox oil pressure	gearbox	18
4520	Tower oscillation X level 1	tower	16
650	Pitch controller communication error	pitch	15
715	Charging circuit pitch	pitch	11

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **Ratings.** Computed at each month-end from data up to that date (latest month-end shown here). AMBER when: the full-load gear-oil pump pressure (median of the last 15 days) is >= 1 bar above the turbine's own baseline (days 60-365 before) AND >= 5 robust standard deviations above the other turbines' rise that month; >= 3 oil-system trip episodes in 6 months; >= 6 pitch trip episodes in 6 months; an ambient-temperature or wind-speed sensor drifting vs the fleet over the last 3 months (1.5 °C, 5%); or a monitored sensor frozen on more than 2% of the last 90 days. RED when the pump rise is >= 2 bar; or a trip rule has held 2 month-ends in a row with >= 3x the amber count and a trip in the last 30 days; or two independent systems are amber at once. Shown as notes but never colour-setting, because they had no reliable lead on events in the validation: low availability (it describes the past), model-based abnormal-day rates (oil, pitch, main bearing, power curve) and missing data. A trip episode merges trips less than 24 h apart.
- **Validation.** Retrospective check on Kelmarsh and Penmanshiel (21 turbines). Thresholds selected on 2017-2020, checked on 2021-2024 (792 turbine-months): pump build-up → oil trip or service stop: 58% of 12 flagged turbine-months vs 55% by chance; repeat oil trips → further oil trips: 63% of 75 flagged turbine-months vs 32% by chance; repeat pitch trips → further pitch trips: 67% of 6 flagged turbine-months vs 36% by chance. Any amber or red: 76% followed by a logged event within 90 days vs 50% by chance (recall 19%); red alone 65% vs 58% on only 17 turbine-months, so red marks strength and persistence more than a higher probability. Held-out ratings: 89% green, 9% amber, 2% red. Consecutive months overlap, so the effective sample is much smaller than the row counts; details in `reports/fleet_proof/calibration.md`.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and a fleet of 4+ turbines (peer comparison). High >= 5, Medium 3-4, Low <= 2.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible. The rating thresholds were calibrated on two UK farms of one turbine family (Senvion MM92/MM82); on other fleets the peer-relative logic still applies but the hit rates are unknown. Ratings look back up to 6 months and only at what SCADA and the alarm log show: inspections, oil analyses, vibration (CMS) data and work orders are not used, so a recent repair or a fault the sensors cannot see changes nothing here.
- **Data.** CC BY 4.0, Cubico Sustainable Investments, doi:10.5281/zenodo.5946808 (H1-2021) / 10.5281/zenodo.16807304.